

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

INTRO TO CRITICAL INFRASTRUCTURE FOR IT LEARNERS

LAB 01 — INTRODUCTION TO PROGRAMMABLE LOGIC CONTROLLERS

Student Edition

Authorized ICDFA training environments only
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Student Name	Pedro Miambo
Registration Number	C11/26/ACIS/17373
Signature	Pedro Miambo
Date	20-09-2026

Lab 01: Introduction to Programmable Logic Controllers

Course: Intro to Critical Infrastructure for IT Learners | **Estimated Duration:** 2–3 hours

Submission deadline: As published in ICDFA E-Campus.

Authorization & Safety: Perform all activities only on the ICDFA-provided virtual machines, packet captures and isolated training networks. Do not bridge these VMs to production networks, scan external systems, or reuse credentials observed in training data.

Learning Outcomes

- Explain the role of PLCs in industrial and critical-infrastructure environments.
- Identify core PLC components including processor, power supply, input modules, output modules, rack/chassis and expansion modules.
- Differentiate discrete (digital) and analog I/O.
- Launch OpenPLC Editor and inspect a ladder-logic motor-control example.
- Explain the purpose of normally open contacts, normally closed contacts and output coils.

Laboratory Environment

System	Expected Address	Account	Password
PLC VM	192.168.1.10	Plc	password

Note: If the instructor-provided VM image uses updated addressing or credentials, use the values published in E-Campus. Do not alter addressing unless instructed.

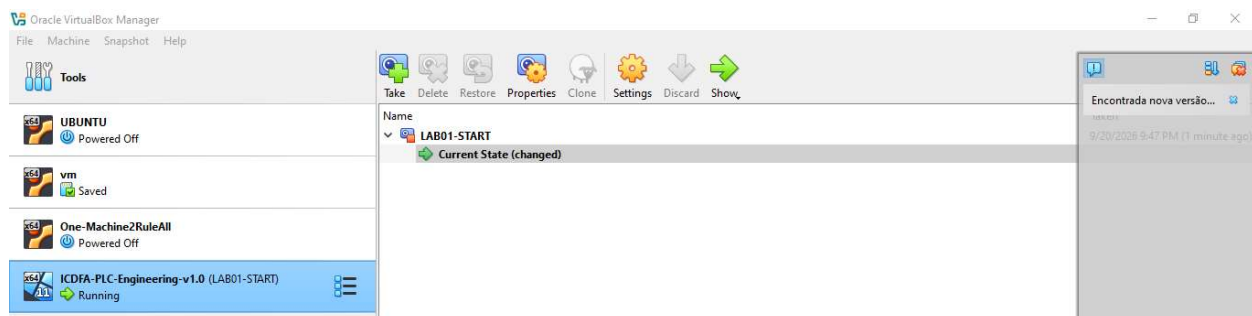
Pre-Lab Checklist

- Read the complete lab before beginning.
- Confirm you can access the required VM(s) and that virtualization support is enabled.
- Use the instructor-specified isolated/host-only network.
- Create or restore the clean starting snapshot for this lab.
- Prepare a report document with your name, registration number, course, lab number and date.
- Capture evidence as you progress rather than repeating the lab at the end.

Procedure

Step 1. Import the ICDFA-provided PLC VM into the virtualization platform specified by your instructor. Configure the laboratory network as Internal/Host-Only unless your instructor provides a different isolated configuration. Do not use Bridged networking.

Step 2. Create a clean snapshot named LAB01-START before making changes. Start the PLC VM and sign in with the assigned laboratory account.



Step 3. Confirm that the VM is using the expected training address 192.168.1.10. If your instructor supplied a revised image or address, record the value in your report instead of changing it without approval.

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INTRO TO CRITICAL INFRASTRUCTURE FOR IT LEARNERS

```
ICDFA-PLC-Engineering-v1.0 (LAB01-START) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

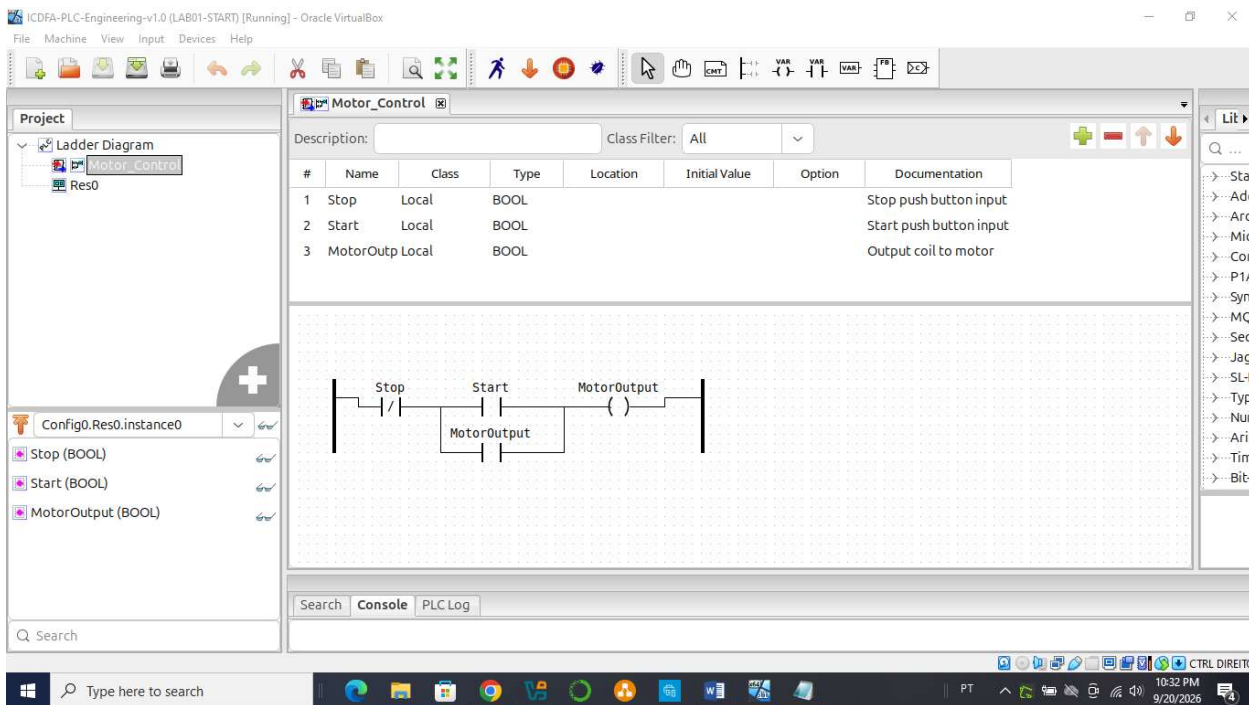
icdfa@icdfa-plc-eng: ~
icdfa@icdfa-plc-eng:~$ ip add
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
   link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
   inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
   inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group
   default qlen 1000
   link/ether 08:00:27:b9:44:42 brd ff:ff:ff:ff:ff:ff
   inet 192.168.56.101/24 brd 192.168.56.255 scope global dynamic noprefixroute
       enp0s3
       valid_lft 483sec preferred_lft 483sec
       inet6 fe80::a00:27ff:feb9:4442/64 scope link
       valid_lft forever preferred_lft forever
icdfa@icdfa-plc-eng:~$
```

VM IP address: 192.168.56.101

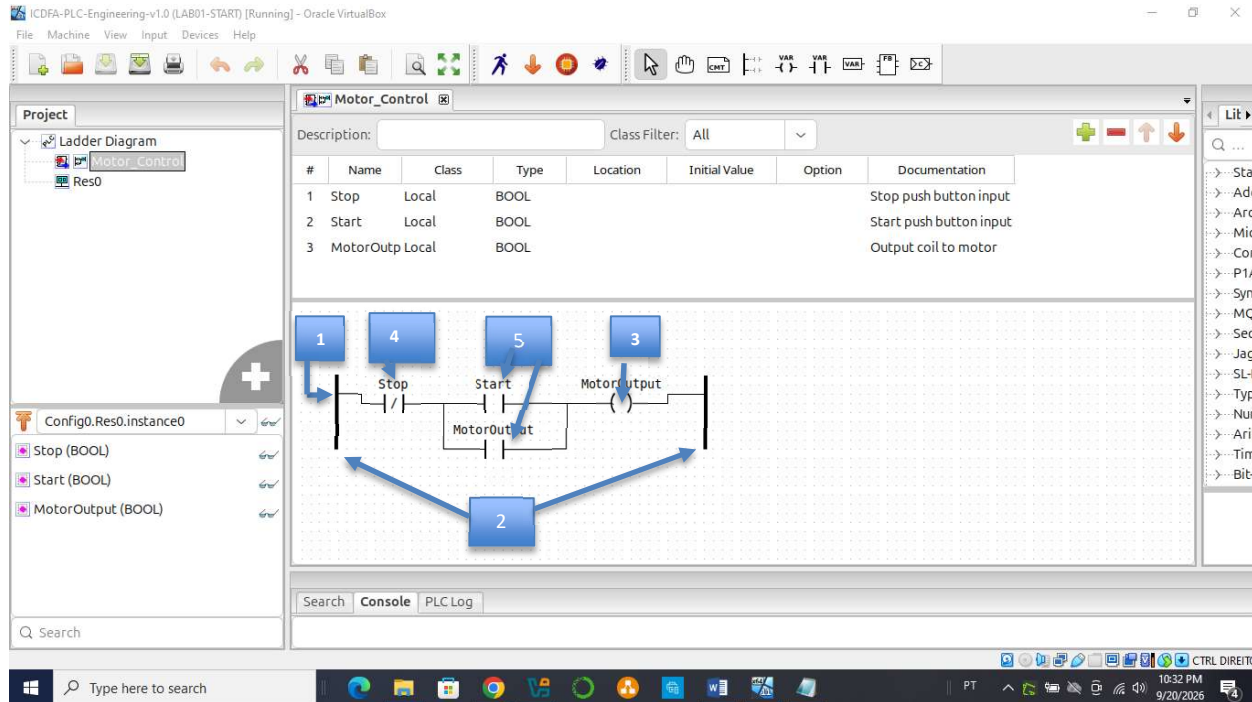
Step 4. Open the OpenPLC Editor from the desktop/application menu.

Step 5. Select File > Open and browse to the provided PLC Programming Language Examples directory. Open the Ladder Diagram project.

Step 6. Under the Project tree, open Res0 and edit instance0 (Motor_Control). Expand the ladder diagram so the full motor-control logic is visible.



Step 7. Identify the two power rails, the rung(s), the Start contact, Stop contact, MotorOutput contact used for latching, and the MotorOutput coil.



- (1) - Rung
- (2) - Power rails
- (3) - MotorOutput Coil
- (4) - Stop Contact
- (5) - Start & MotorOutput contacts

Step 8. Trace the logic from the left rail to the right rail. Explain what must be true for the motor output to energize and what causes it to de-energize.

A: The circuit works the following way:

- ✓ When Operator presses **Start**, the contact closes. Since the **Stop** button is already closed, electric current travels across the rung from the left rail to the right rail, energizing the **MotorOutput** coil.
- ✓ When Operator presses **Stop**, that contact opens up. This completely cuts off the power pathway. The motor coil loses power, which causes the parallel latching contact to snap back open. Even when the **Stop** button is released, the circuit remains dead until **Start** is pressed again.

Step 9. Record one example each of a discrete input, analog input, discrete output and analog output that could appear in an industrial process.

A: Following examples appear in industrial process:

I/O Classification	Examples
Discrete input	Physical emergency stop button or a proximity sensor
Analog input	Temperature sensor, or a pressure transmitter
Discrete output	A warning strobe light, or a basic open/close water valve
Analog output	A control signal for a flow valve, or a control signal to adjust motor speed

Step 10. Capture the required evidence, save your report, then restore or power off the VM as directed by the instructor.

Required Evidence & Submission

- Screenshot showing the OpenPLC Motor_Control ladder diagram.
- A labelled explanation of the Start, Stop, latch and MotorOutput elements.
- A completed I/O classification table with at least four industrial examples.
- A 150–250 word reflection on why network-connected PLCs create both operational benefits and cybersecurity considerations.

Recommended filename: <RegistrationNo>_LAB01_Introduction_to_Programmable_Logic_Controllers.pdf

Knowledge Check

1. What is the difference between an input module and an output module?

Response: An input module listens to the machine, while an output module talks to the machine. In a Programmable Logic Controller (PLC), these modules act as the eyes, ears, and hands of the system, allowing the central computer (CPU) to interact with the real world.

2. Why is a normally closed contact commonly associated with stop or emergency-stop logic?

Response: We use Normally Closed contacts for emergency stops to ensure the system is fail-safe. If anything goes wrong—like a broken wire, a component failure, or a power outage—the circuit drops out and immediately stops the machine.

3. Give one example of a discrete I/O signal and one example of an analog I/O signal.

Response:

Classification	Examples
Discrete I/O	Push button
Analog input I/O	Temperature sensor

4. What operational risk can arise when industrial automation devices become network-connected?

Response: Connecting PLCs to networks helps bridge the gap between operational technology and information technology. This allows operators to efficiently monitor field devices and act in a timely manner; however, this advantage comes at the cost of an increased attack surface. Operational risks derived from connecting industrial automation devices to network include:

- ✓ Many older PLCs do not support security features such as encryption, user authentication, or secure protocols.
- ✓ Connecting a PLC to a local network, or the internet, creates a path for threat actors. A vulnerable engineering workstation can allow physical control of a pipeline or a power grid.
- ✓ Traditional cyberattacks steal data, but a hacked PLC can cause actual physical damage, environmental disasters, product contamination, or even threaten human lives

Marking Rubric — 100 Marks

Criterion	Marks	Standard
Technical execution	50	Lab procedure completed correctly; required configuration/analysis works as intended.
Evidence	20	Screenshots/files are complete, readable, relevant and clearly labelled.
Analysis & knowledge check	20	Responses demonstrate understanding of the industrial-control concepts observed.
Professional reporting	10	Clear structure, correct terminology, concise writing, file naming and submission compliance.

Submission Integrity: Evidence must come from the student's own assigned laboratory environment. Screenshots should show enough context to verify the activity. Do not submit another learner's screenshots or project files.